
Course Assessment Plan — E-Commerce & Web Security

This course uses a combination of quizzes, practical assignments, and a group case study to ensure students gain theoretical understanding **and** hands-on skills in modern web security and e-commerce technologies.

Quiz & Assignment Breakdown

◆ Quiz 1 — Security Foundations (Marks: 15)

Topics Covered:

- Security Concepts
- E-Commerce Security Solutions
- SQL Injection
- XSS

Focus: Core vulnerabilities and traditional security mechanisms.

◆ Quiz 2 — System Design + Practical SQL Assignment (Marks: 15)

Group Work: 2 students per group

Part A: Proposal of a New E-Commerce System (6 Marks)

Students will design and propose a new e-commerce system covering:

- Business Models
- Online Presence
- Marketing Strategy
- Digital Marketing Plan
- Modern Security Guidelines

A mix of business planning and secure system design.

Part B: Practical SQL Injection Attack & Protection (9 Marks)

Hands-on demonstration focusing on:

- Crafting SQL injection payloads
 - Detecting vulnerable endpoints
 - Applying secure coding & parameterized queries
 - Showing the before/after security effects
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◆ Quiz 3 — E-Commerce & Modern Security Landscape (Marks: 15)

Topics Covered:

- Lecture-8-(Chapter 4) Building an E-Commerce Presence & Digital Marketing Concepts
- Lecture-9 XSS and CSRF
- Lecture-10 Overview on Vulnerability Assessment and Penetration Testing
- Lecture-11 Modern Web Security_ Threats, Defenses

Focus: Business + Technology + Modern Security Practices.

◆ Quiz 4 — Practical Skills Demonstration (Marks: 15)

Group Work: 2 students per group

This assessment includes **one practical components**:

Open-Ended Practical Assignment (Choose ANY 1 Task)

Goal:

Provide hands-on experience in core web security vulnerabilities **and** applied e-commerce technologies, including modern AI-based tools.

Students must select ONE task from the list below:

1. XSS Attack & Protection

- Demonstrate attack
- Show protection

2. CSRF Attack & Protection

- Demonstrate attack
- Show protection

3. SEO (Search Engine Optimization) & Web Analytics Demo

- Optimize a webpage and analyze SEO performance
- Integrate and interpret analytics dashboards

4. Sniffing / Eavesdropping + WAF Protection Demo

- Capture insecure traffic & show how HTTPS/WAF mitigates threats

5. Web Vulnerability Test / Scan

- Use tools like ZAP/Burp/Nikto
- Report vulnerabilities & mitigations

6. Use of a Trained AI Model to Secure a Website

- Apply a pre-trained phishing/URL/XSS/SQLi detection model
- Integrate it into a simple security workflow

Assessment	Marks	Type / Components	Key Topics / Activities	Group?	Submission Deadline
Quiz 1 — Security Foundations	15	Short Questions	• Security Concepts• E-Commerce Security Solutions• SQL Injection• XSS & CSRF• Password Reset Poisoning	No	Week 6
Quiz 2 — System Design + SQL Practical	15	Part A: SQL Injection Practical Part B: New E-Commerce System Proposal	Part A: • SQL Injection Attack Demo• Secure Coding (Parameterized Queries) Part B: • Business Model Design• Online Presence Plan• Marketing & Digital Strategy• Security Guidelines	Yes (2 per group)	Part A: Week 5 Part B: Week ?
Quiz 3 — E-Commerce & Modern Security Landscape	15	Short Questions	• Building an E-Commerce Presence• Digital Marketing Concepts• E-Commerce Crime & Security Threats• Modern Web Security: Threats & Defenses	No	Week 13
Quiz 4 — Practical Skills Demonstration	15+ (Bonus)	Open-Ended Practical + Presentation	Available Tasks: (Choose any ONE Task) 1. XSS & CSRF Attack + Protection 2. CSRF Attack + Protection 3. SEO Demo & Web Analytics Demo 4. Sniffing/Eavesdropping + WAF Demo 5. Web Vulnerability Scan 6. Use of Pre-Trained AI Model for Security	Yes (2 per group)	Week 14

Overall Learning Goal

Provide students with practical and analytical experience in:

- Core web vulnerabilities and defenses
- E-commerce system design and business modeling
- Digital marketing and online presence strategies
- Real security testing tools and AI-based defenses

This ensures a **balanced, industry-aligned understanding** of both e-commerce technologies and modern web security practices.
